

Stacking Ensemble Model Raporu

General Panel - In-Silico Skorlar OLMADAN (v3_no_insil)

Tarih: 2026-05-08 00:33

Baz Modeller: LIGHTGBM, XGBOOST, DNN, SVM

OOF Fold Sayisi: 5 | Meta-feature: 1x8 vektor (4 x [p_benign, p_patho])

Meta Modeller: LightGBM, NN | Diversity Meta: LightGBM(19), NN(19)

1. Yontem Ozeti

Veri seti : ClinVar General Panel - 3156 variant (pos=2182, neg=974)

Bolunme : %80 CV (2524 satir) + %20 hold-out (632 satir), stratified

ADIM 1 - OOF Baz Model Egitimi:

5-fold StratifiedKFold; her fold icin grid search yapilir.

Her modelin OOF skoru temiz meta-feature matrisini olusturur (data leakage yok).

Final model tum X_cv ile yeniden egitilir; hold-out skorlari uretilir.

ADIM 2 - Standart Meta-Feature (1x8):

Her baz model icin [p_benign, p_pathogenic] => 4 model x 2 = 8 sutun.

ADIM 3 - Diversity-Aware Meta-Feature (1x19):

8 standart sutun + 11 cesitlilik sutunu:

disagreement(std), confidence(ort), entropy, max_min_gap,

diff_lgbm_xgb, diff_lgbm_dnn, diff_lgbm_svm,

diff_xgb_dnn, diff_xgb_svm, diff_dnn_svm.

ADIM 4 - Meta-Model Egitimi:

Standart (8 sutun): LightGBM + NN, 3-fold CV ile en iyi hyperparametre secilir.

Diversity (19 sutun): LightGBM + NN, ayni prosedur, NN hidden1=64 hidden2=32.

2. Baz Model Hold-Out Metrikleri

Model	F1	AUC-ROC	AUC-PR	Precision	Recall	MCC
lightgbm	0.9454	0.9715	0.9862	0.9217	0.9703	0.8153
xgboost	0.9451	0.9682	0.9851	0.9254	0.9657	0.8154
dnn	0.8850	0.8904	0.9407	0.8487	0.9245	0.5948
svm	0.8955	0.8960	0.9436	0.8623	0.9314	0.6352

(Ek metrikler: *Balanced Accuracy / Cohen Kappa / Threshold*)

Model	Bal.Acc	Cohen's K	Threshold
lightgbm	0.8928	0.8122	nan
xgboost	0.8957	0.8133	nan
dnn	0.7776	0.5878	nan
svm	0.7990	0.6293	nan

3. Standart Meta-Model Metrikleri

Model	F1	AUC-ROC	AUC-PR	Precision	Recall	MCC
stacking_lgbm	0.9444	0.9620	0.9754	0.9179	0.9725	0.8116
stacking_nn	0.9505	0.9664	0.9817	0.9357	0.9657	0.8347

(Ek metrikler: *Balanced Accuracy / Cohen Kappa / Threshold*)

Model	Bal.Acc	Cohen's K	Threshold
stacking_lgbm	0.8888	0.8075	0.280
stacking_nn	0.9085	0.8335	0.170

4. Diversity-Aware Meta-Model Metrikleri

Model	F1	AUC-ROC	AUC-PR	Precision	Recall	MCC
diversity_lgbm	0.9445	0.9606	0.9755	0.9350	0.9542	0.8164
diversity_nn	0.9495	0.9661	0.9812	0.9517	0.9474	0.8373

(Ek metrikler: *Balanced Accuracy / Cohen Kappa / Threshold*)

Model	Bal.Acc	Cohen's K	Threshold
diversity_lgbm	0.9028	0.8159	0.550
diversity_nn	0.9198	0.8373	0.520

5. Karsilastirma Ozeti

En iyi baz model : lightgbm F1=0.9454 AUC=0.9715 MCC=0.8153
En iyi meta model : stacking_nn F1=0.9505 AUC=0.9664 MCC=0.8347
En iyi div. model : diversity_nn F1=0.9495 AUC=0.9661 MCC=0.8373

Genel en iyi : stacking_nn F1=0.9505

Stacking kazanimi (meta vs baz) : dF1 = +0.0051

Diversity kazanimi (div vs baz): dF1 = +0.0042

6. Diversity Feature Aciklamasi

disagreement : 4 baz modelin pos-skor std sapmasi (modeller ne kadar ayrisiyor)

confidence : 4 baz modelin pos-skor ortalamasi (ensemble guven skoru)

entropy : binary entropi; 0.5 civarinda max belirsizlik

max_min_gap : max(pos_scores) - min(pos_scores)

diff_lgbm_xgb : |LightGBM_pos - XGBoost_pos|

diff_lgbm_dnn : |LightGBM_pos - DNN_pos|

diff_lgbm_svm : |LightGBM_pos - SVM_pos|

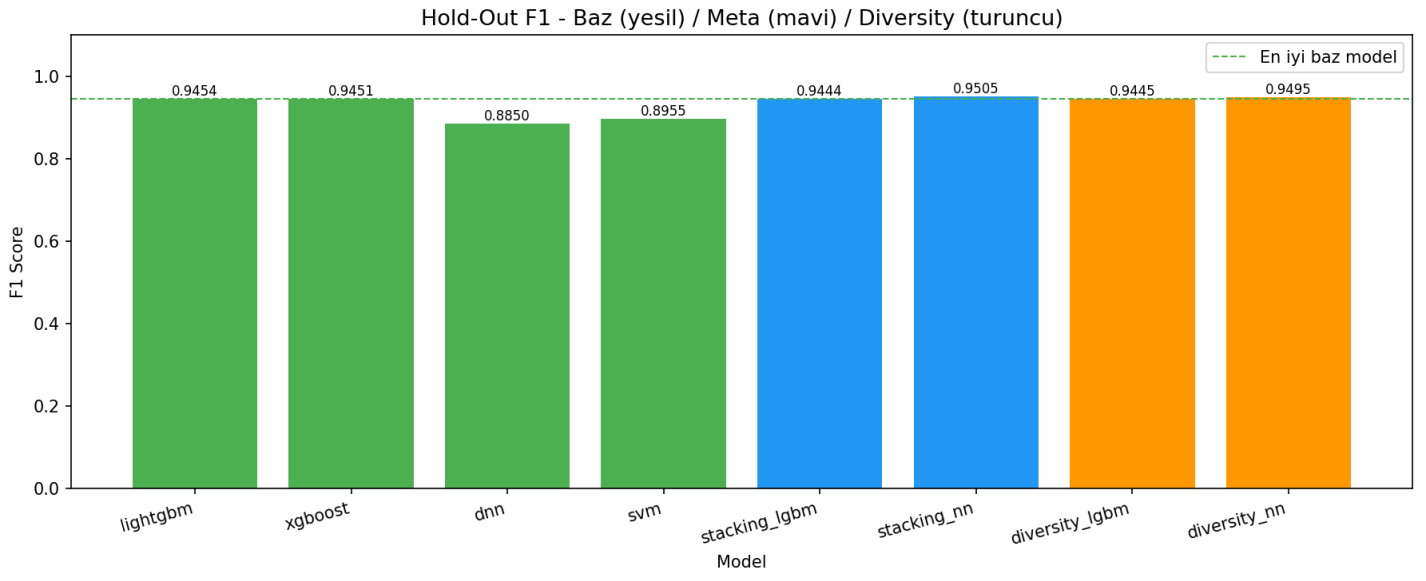
diff_xgb_dnn : |XGBoost_pos - DNN_pos|

diff_xgb_svm : |XGBoost_pos - SVM_pos|

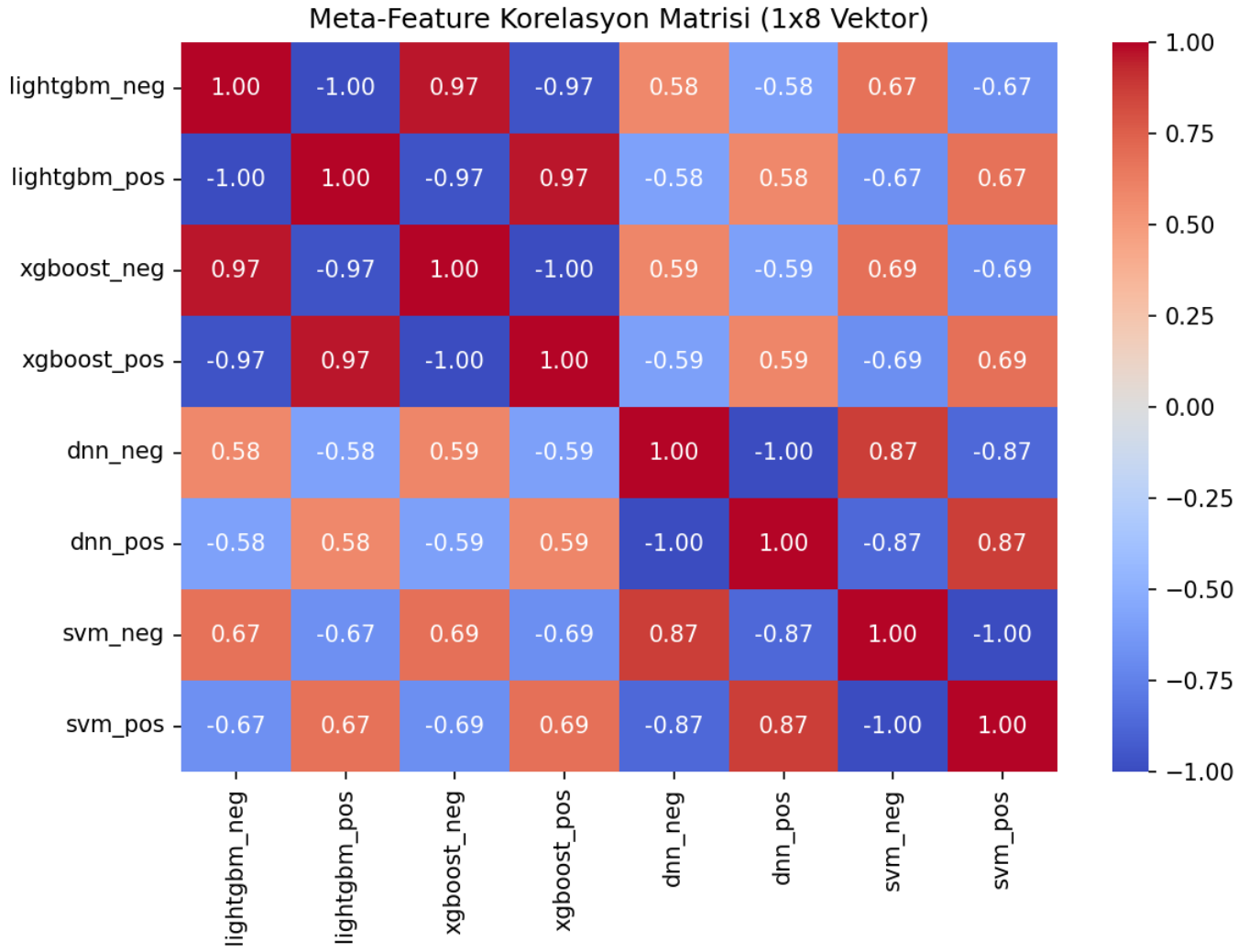
diff_dnn_svm : |DNN_pos - SVM_pos|

Toplam: 8 (orijinal) + 11 (diversity) = 19 meta-feature sutun

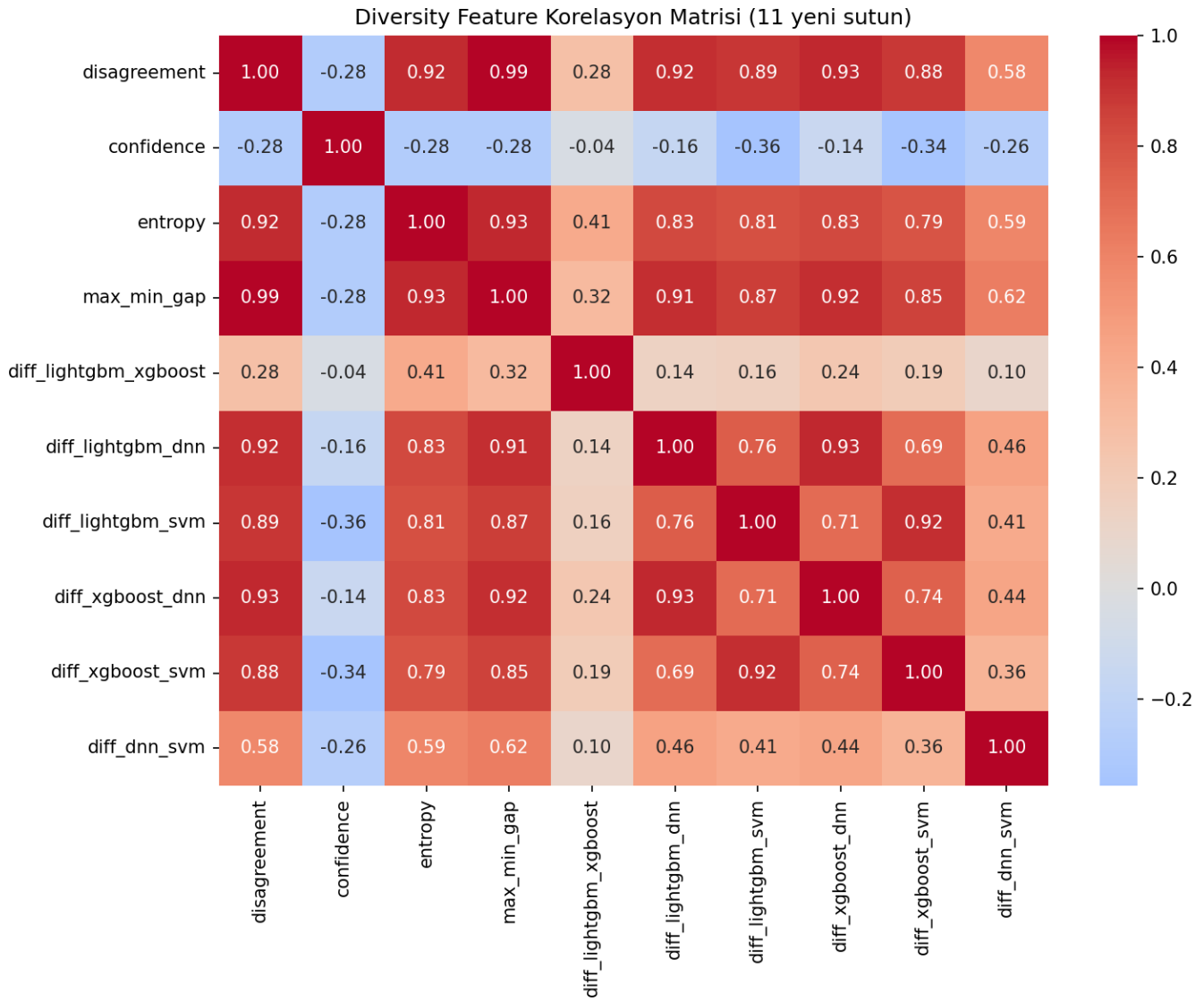
F1 Comparison



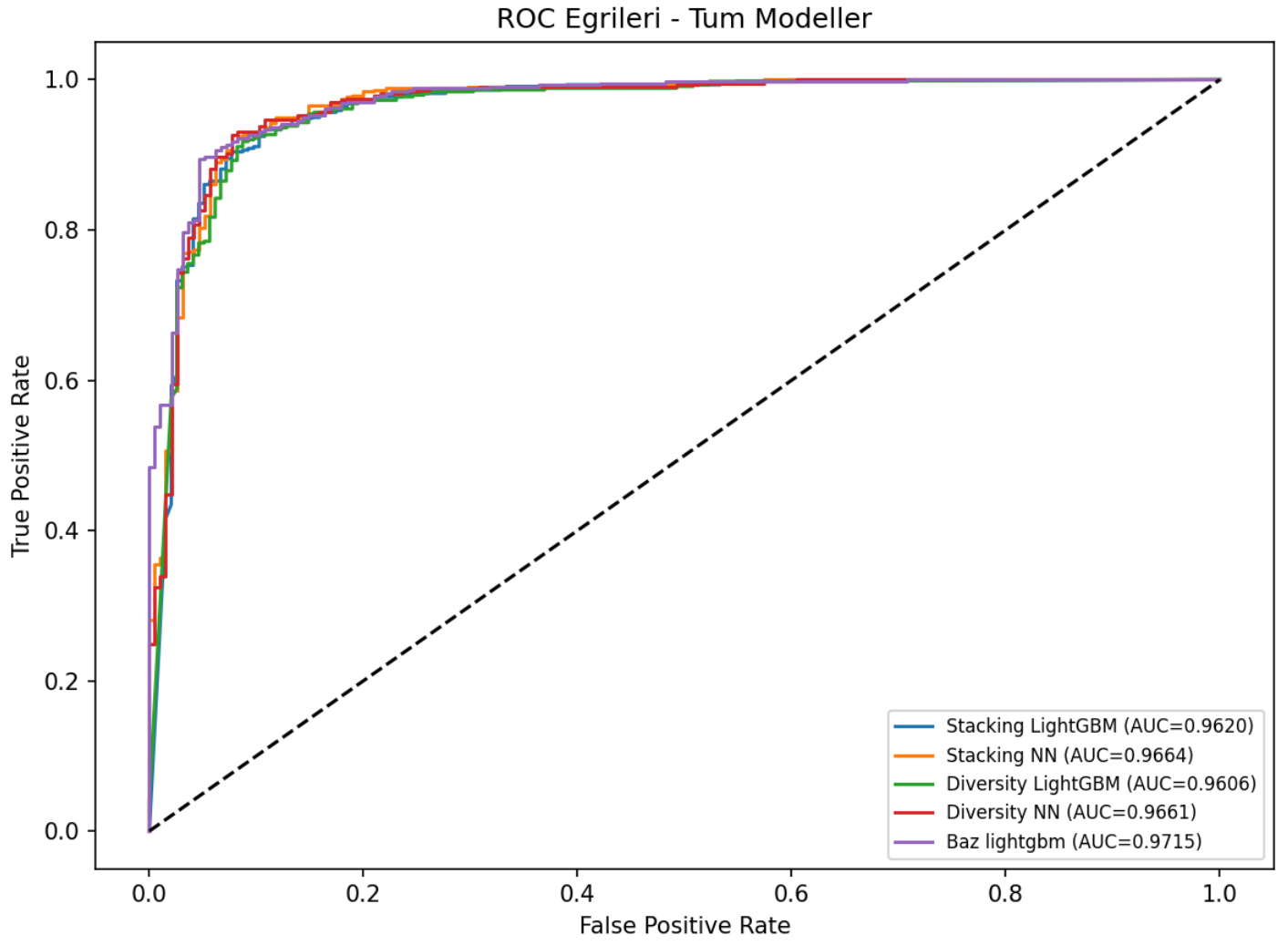
Meta Feature Correlation



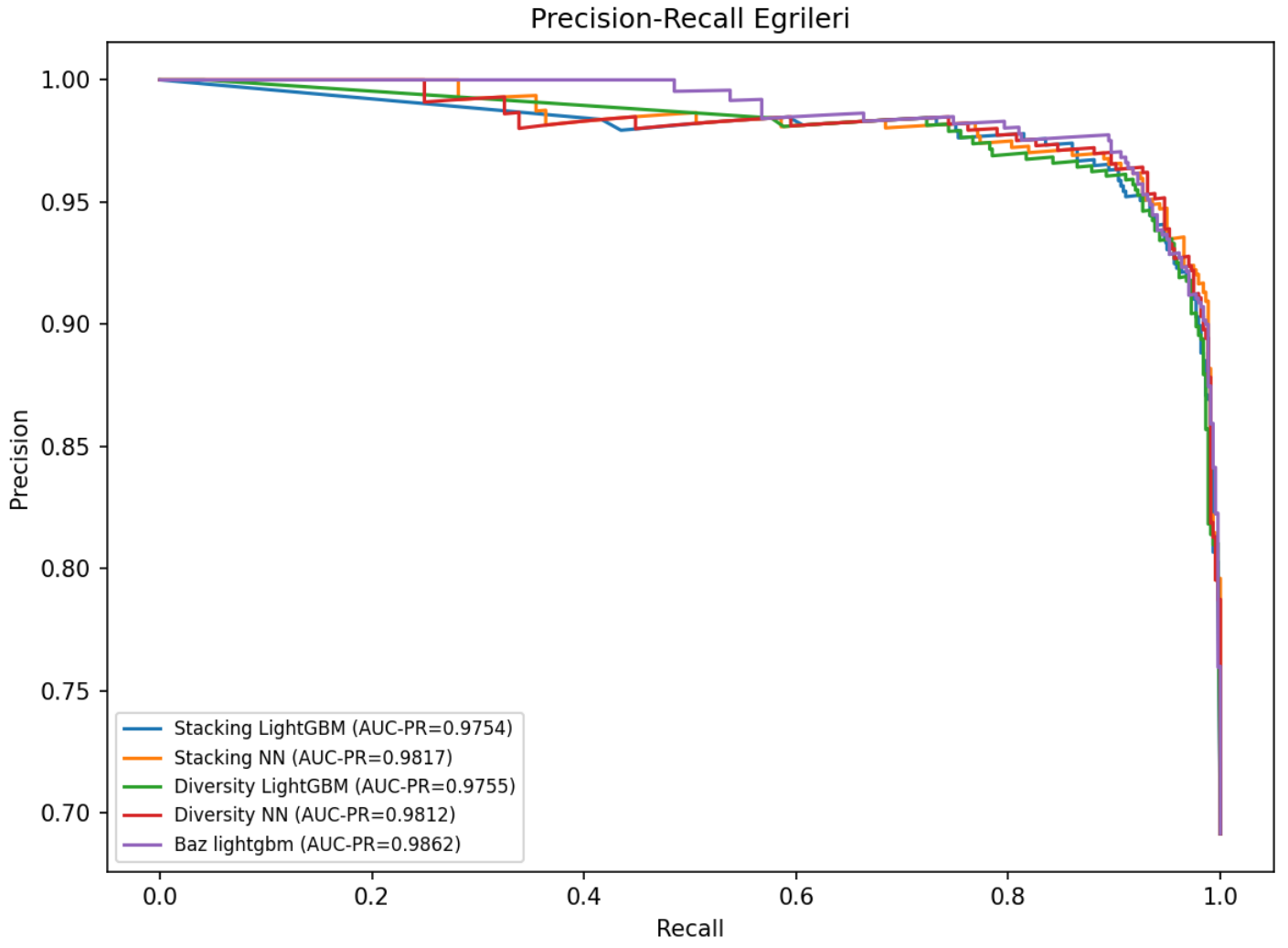
Diversity Feature Correlation



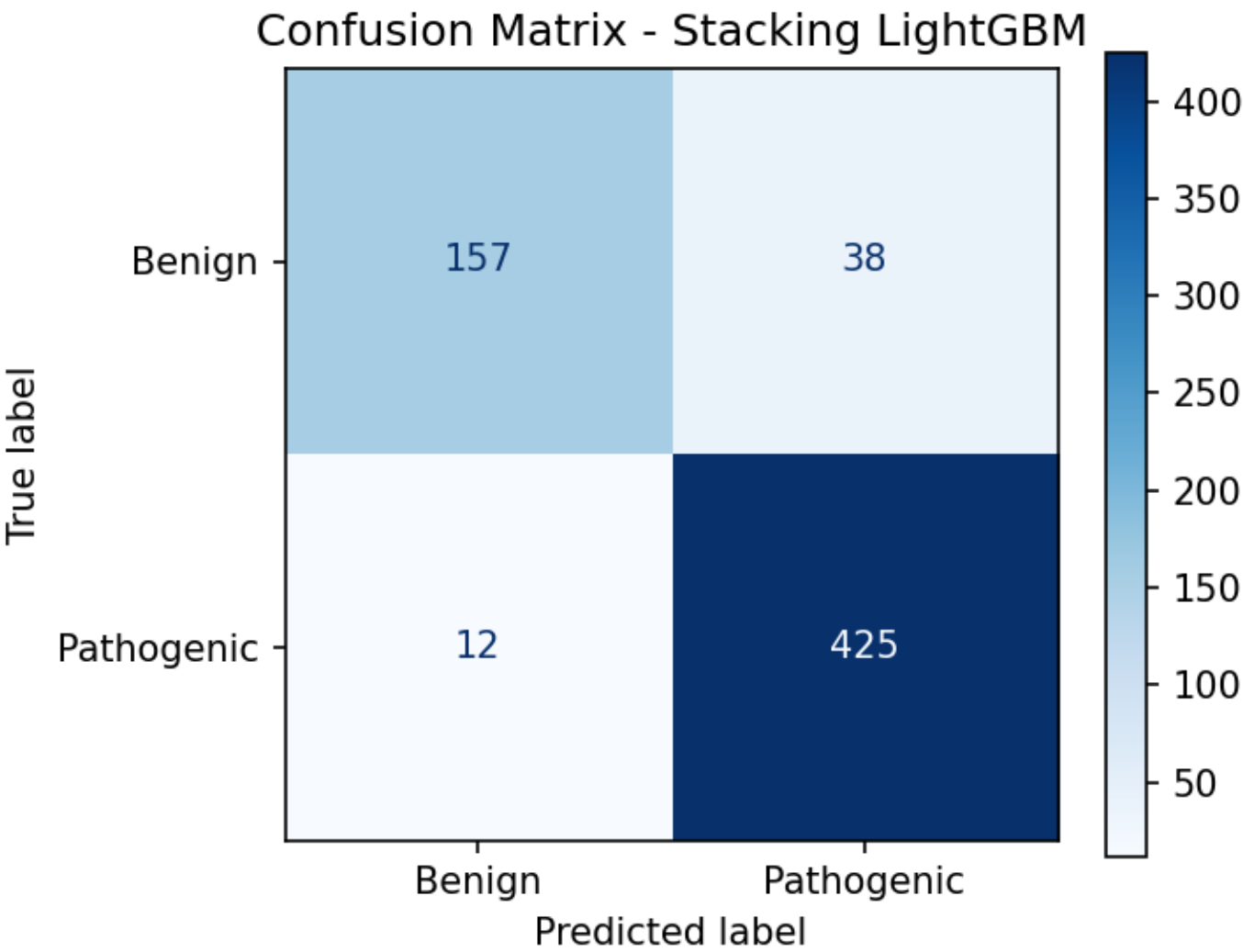
Roc Curves



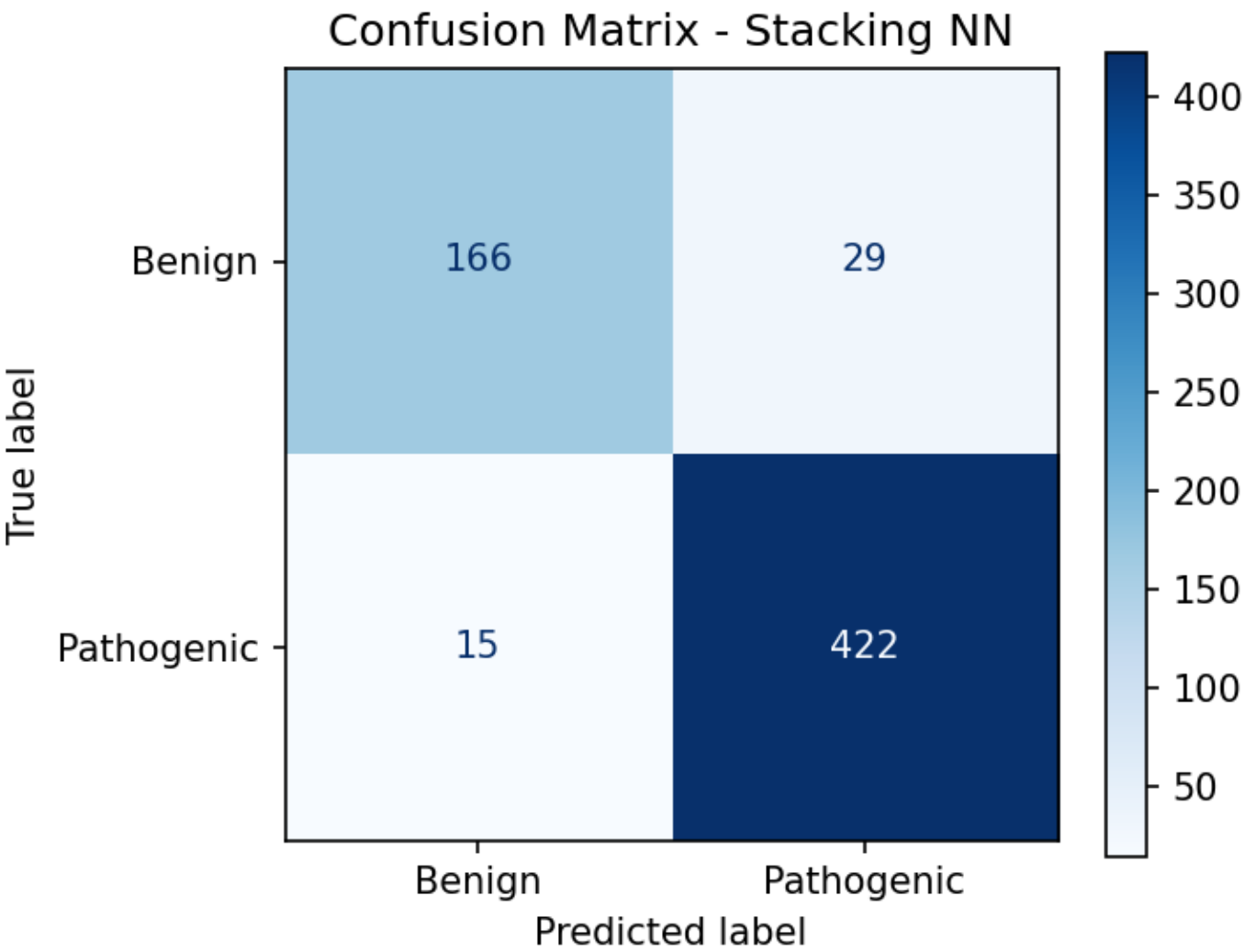
Pr Curves



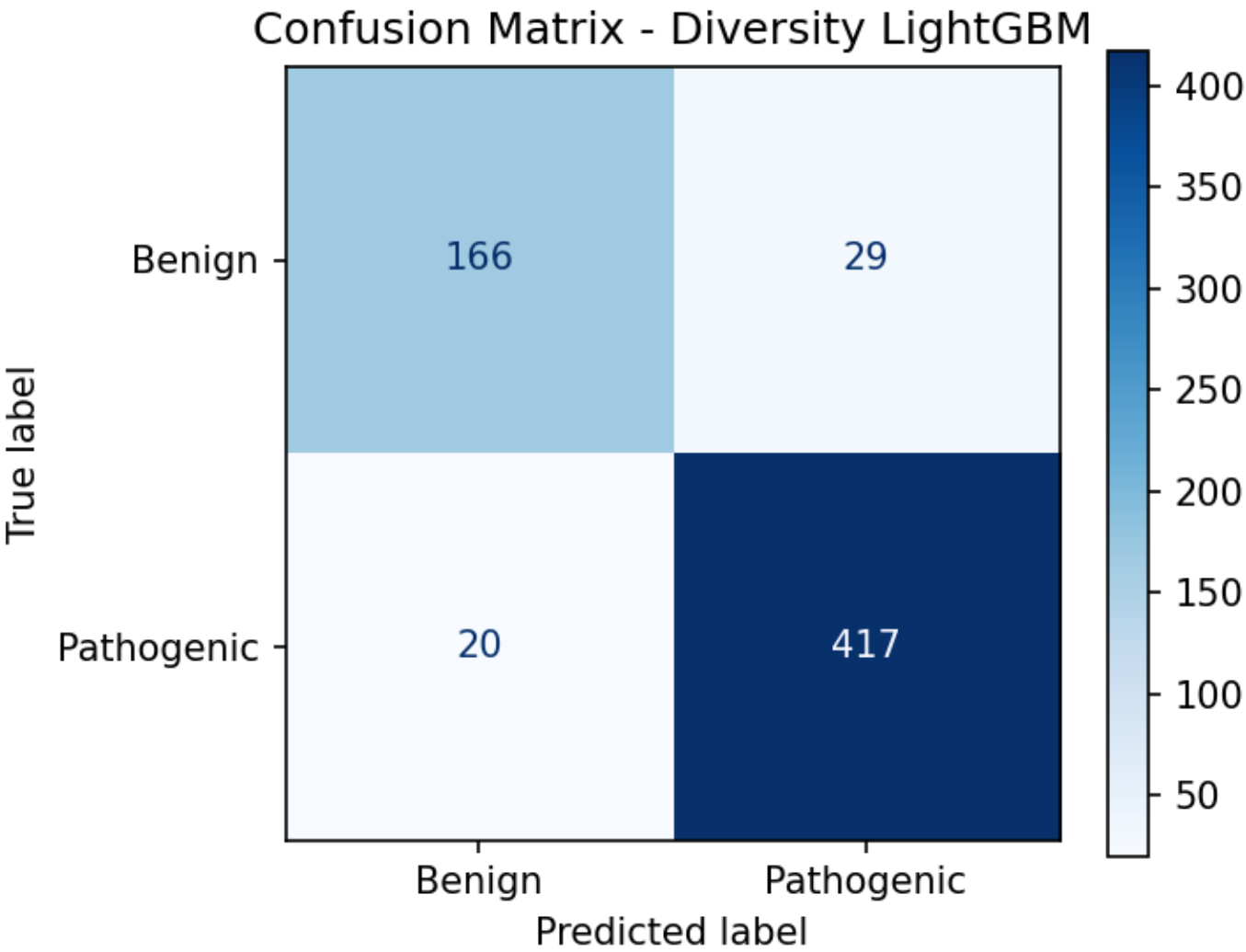
Stacking Lightgbm Confusion Matrix



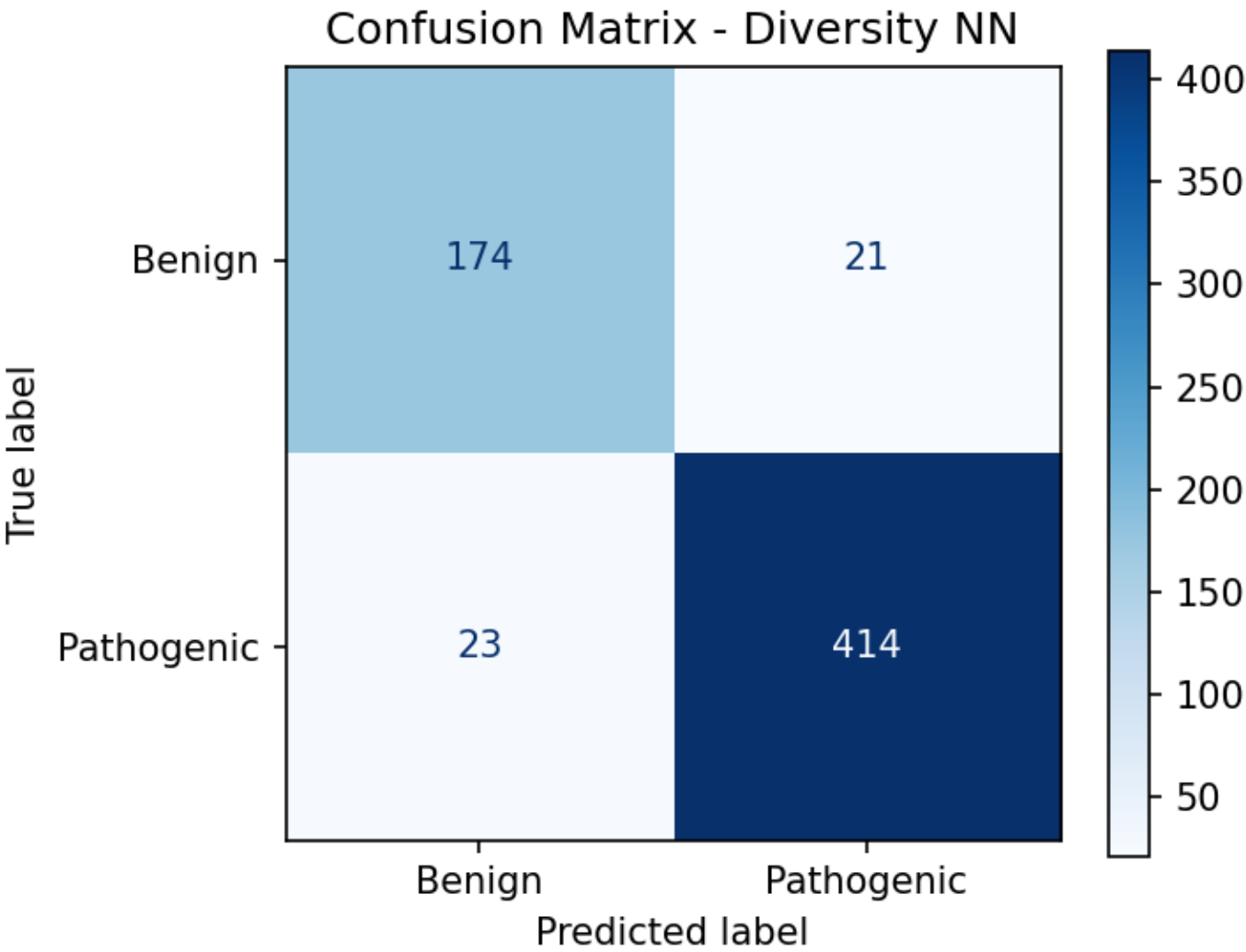
Stacking Nn Confusion Matrix



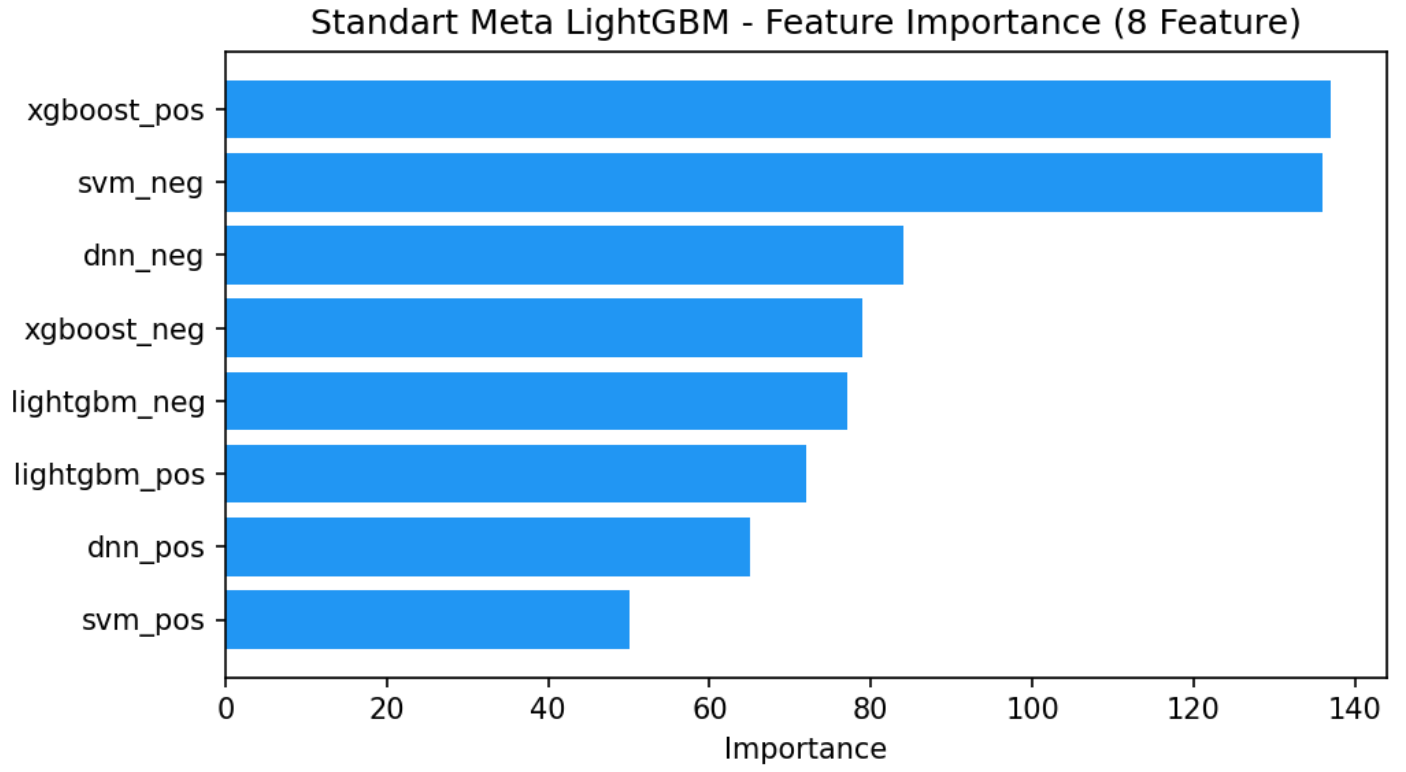
Diversity Lightgbm Confusion Matrix



Diversity Nn Confusion Matrix



Meta Lgbm Feature Importance



Diversity Lgbm Feature Importance

